Project Executable Files

OptiCrop — Smart Agricultural Production Optimization Engine

How to Run the Application

- 1. Clone the repository: git clone https://github.com/neelimavana/OptiCrop.git
- 2. Navigate into the project folder: cd OptiCrop
- 3. Create and activate a virtual environment: python -m venv venv, then venv\Scripts\activate (Windows)
- 4. Install dependencies: pip install -r requirements.txt
- 5. Run the application: python app.py
- 6. Open the printed URL (http://127.0.0.1:5000) in a browser.

Key Executable Files

- app.py — the main Flask application entry point; running this starts the web server.
- notebooks/EDA_and_Model_Training.ipynb — reproduces the full ML pipeline from raw data to saved model artifacts (optional — pre-trained artifacts are already included in model/).

Pre-Trained Artifacts (no retraining required to run the app)

- model/model.pkl — trained Random Forest classifier (99.55% test accuracy).
- model/scaler.pkl — StandardScaler fitted on the training data.
- model/label_encoder.pkl — maps numeric predictions back to crop names.